

HEART DISEASE ANALYSIS

PROJECT TEAM MEMBERS

Yaswanth Mutyala-Team Lead

Likitha Satti- member

Kovvuri Jyothsna Harini- member

Varshini Siddireddy- Member

Manukonda Nikhitha Florence -member

1. INTRODUCTION

1.1 Project Overview

1.2 Purpose

2. IDEATION PHASE

2.1 Problem Statement

2.2 Empathy Map Canvas

2.3 Brainstroming

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

3.2 Solution Requirement

3.3 Data Flow Diagram

3.4 Technology Stack

4. PROJECT DESIGN

4.1 Problem Solution Fit

4.2 Proposed Solution

4.3 Solution Architecture

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Functional Testing

6.2 Performance Testing

7. RESULTS

7.1 Output Screenshots

8. ADVANTAGES & DISADVANTAGES

9. CONCLUSION

10. FUTURE SCOPE

11. APPENDIX

INTRODUCTION

Heart disease is one of the leading causes of death worldwide and remains a major public health concern. The increasing prevalence of unhealthy lifestyles, poor dietary habits, lack of physical activity, smoking, obesity, and stress has significantly contributed to the rise in cardiovascular diseases. Early detection of heart disease and timely identification of risk factors are essential for reducing complications and improving patient outcomes. Since healthcare institutions generate large volumes of patient data, manual analysis becomes difficult and time-consuming, making data-driven approaches increasingly important.

This project, "**Heart Disease Diagnostic Dashboard using MySQL, Tableau, and Flask**," aims to analyze heart disease data and transform it into meaningful visual insights. The patient dataset is stored in a **MySQL** database, where SQL operations are used for efficient data management and retrieval. The processed data is then connected to **Tableau Public** to create interactive charts, dashboards, and stories that highlight important clinical indicators such as age, gender, cholesterol levels, chest pain type, blood pressure, exercise-induced angina, and heart disease status. Finally, the Tableau dashboard is embedded into a **Flask** web application, providing users with an easy-to-use web interface for exploring the analysis.

The interactive dashboard enables healthcare professionals, researchers, and policymakers to identify disease patterns, compare patient groups, and understand the relationship between various health factors. By presenting complex medical data through clear visualizations, the project supports informed decision-making and promotes preventive healthcare. Overall, this project demonstrates how database management, business intelligence, and web technologies can be integrated to build an effective healthcare analytics solution that enhances awareness and facilitates data-driven decision-making.

1.1 PROJECT OVERVIEW

The **Heart Disease Diagnostic Dashboard** is a healthcare analytics project developed to analyze and visualize heart disease patient data using **MySQL**, **Tableau**, and **Flask**. The project focuses on transforming raw medical data into meaningful and interactive visualizations that help identify key risk factors associated with heart disease. Patient information such as age, gender, chest pain type, cholesterol levels, resting blood pressure, maximum heart rate, exercise-induced angina, and heart disease status is stored in a MySQL database and analyzed using SQL queries. The processed data is then connected to Tableau Public to create interactive dashboards and stories that provide a clear understanding of patient health trends.

The project integrates the Tableau dashboard into a **Flask** web application, allowing users to access the visualizations through a simple and user-friendly interface. Interactive filters, charts, and dashboards enable users to compare patient groups, analyze disease patterns, and gain valuable insights for healthcare decision-making.

1.2 PURPOSE

The primary purpose of this project is to analyze heart disease patient data and present meaningful insights through interactive dashboards using **MySQL**, **Tableau**, and **Flask**. The project aims to transform raw healthcare data into visually appealing reports that help identify important risk factors such as age, gender, cholesterol levels, blood pressure, chest pain type, exercise-induced angina, and heart disease status. By organizing and visualizing the data effectively, the project supports better understanding of disease patterns and enables users to make informed decisions based on the analysis.

Additionally, the project demonstrates the integration of database management, business intelligence, and web technologies to develop a complete healthcare analytics solution. It provides healthcare professionals, researchers, and policymakers with an easy-to-use platform for exploring patient data, comparing health indicators, and identifying high-risk groups. The interactive dashboard also promotes awareness about heart disease and highlights the importance of data-driven decision-making in improving preventive healthcare and patient outcomes.

2. IDEATION PHASE

The ideation phase focuses on identifying a real-world healthcare problem and designing an effective solution using data analytics and visualization techniques. Heart disease continues to be one of the leading causes of death worldwide, and healthcare institutions generate vast amounts of patient data that are often difficult to analyze using traditional methods. The need for an interactive system capable of transforming raw medical data into meaningful insights led to the development of this project. By leveraging modern technologies such as **MySQL**, **Tableau**, and **Flask**, the project aims to simplify the analysis of heart disease data and present it in a visually intuitive manner.

The proposed solution is to develop a **Heart Disease Diagnostic Dashboard** that enables users to analyze patient information through interactive charts, dashboards, and stories. The system helps identify key risk factors, compare patient characteristics, and visualize trends related to heart disease. This approach supports healthcare professionals, researchers, and policymakers in making informed decisions, improving disease awareness, and encouraging preventive healthcare through data-driven insights.

2.1 Problem Statement

Heart disease is one of the leading causes of mortality worldwide, affecting millions of people each year. Healthcare organizations collect large volumes of patient data, including demographic information, medical history, lifestyle habits, and clinical indicators. However, analyzing this data manually is time-consuming and often fails to reveal meaningful patterns that can support early diagnosis and preventive care. The lack of effective visualization tools makes it difficult for healthcare professionals to identify high-risk patients and make informed decisions.

The primary problem addressed in this project is the need for an interactive and efficient system that can analyze heart disease data and present it in an easy-to-understand format. By integrating **MySQL** for data management, **Tableau** for interactive data visualization, and **Flask** for web-based access, the project transforms raw healthcare data into meaningful dashboards and stories. This enables users to identify key risk factors, analyze patient trends, and support data-driven decision-making for improved healthcare.

2.2 Empathy Map Canvas

User: Healthcare Professionals (Doctors, Cardiologists, Hospital Administrators, and Patients)

Says

- "I need a quick and accurate way to analyze patient data."
- "It is difficult to identify high-risk patients from large datasets."
- "Interactive dashboards can simplify complex medical information."
- "I want to make informed healthcare decisions using reliable data."

Thinks

- Early diagnosis can reduce the risk of severe heart disease.
- Visualizing patient data helps identify important health trends.
- Data-driven insights improve treatment planning and preventive care.
- A user-friendly system will save time and improve efficiency.

Does

- Reviews patient medical records and clinical reports.
- Analyzes cholesterol, blood pressure, age, and other health indicators.
- Uses dashboards to compare patient groups and identify risk factors.
- Monitors trends to support healthcare decisions and awareness programs.

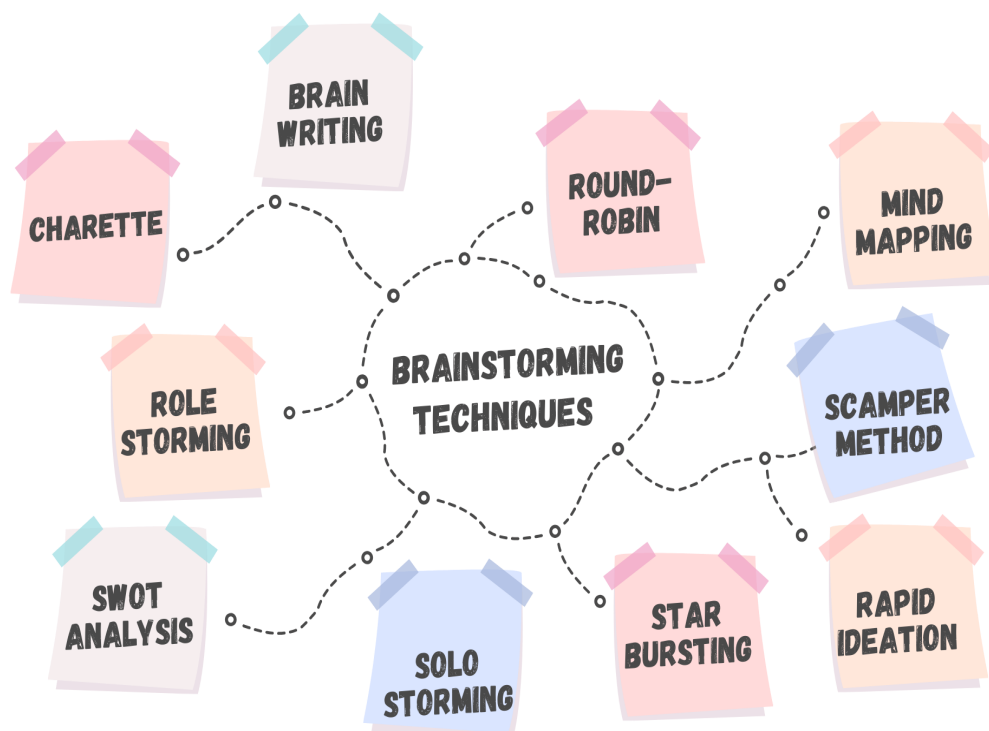
Feels

- Concerned about the increasing number of heart disease cases.
- Needs faster and more effective methods for analyzing patient data.
- Confident when decisions are supported by clear visualizations.
- Motivated to improve patient outcomes through preventive healthcare.

2.3 Brainstorming

The brainstorming phase involved identifying the most effective approach to analyze and present heart disease data in a meaningful way. Different ideas were explored, including the use of spreadsheets, static reports, and data visualization tools. After evaluating these options, it was decided to develop an interactive dashboard using **MySQL**, **Tableau**, and **Flask**. MySQL was selected for efficient data storage and SQL-based data management, Tableau was chosen for its powerful interactive visualization capabilities, and Flask was used to integrate the dashboard into a web application for easy accessibility.

During this phase, the team identified the key health indicators required for analysis, such as **age**, **gender**, **chest pain type**, **cholesterol levels**, **resting blood pressure**, **maximum heart rate**, **exercise-induced angina**, **ST slope**, and **heart disease status**. Various visualization techniques, including bar charts, pie charts, scatter plots, and dashboards, were considered to effectively represent the data. The brainstorming process helped define the project workflow, select appropriate technologies, and finalize a solution that provides clear, interactive, and data-driven insights to support healthcare professionals in understanding heart disease risk factors and trends.



3. REQUIREMENT ANALYSIS

The **Requirement Analysis** phase focuses on identifying the functional and non-functional requirements needed to successfully develop the **Heart Disease Diagnostic Dashboard**. The project requires a reliable heart disease dataset containing patient information such as age, gender, chest pain type, cholesterol levels, resting blood pressure, maximum heart rate, exercise-induced angina, ST slope, and heart disease status. The data is stored in a **MySQL** database, where SQL operations are performed for efficient data management, retrieval, and analysis. **Tableau Public** is used to create interactive visualizations, dashboards, and stories, while **Flask** is used to integrate the Tableau dashboard into a web application, providing users with a simple and accessible interface.

The system is designed to enable users to analyze patient data through interactive charts, apply filters to compare different patient groups, and identify trends and risk factors associated with heart disease. The application should be user-friendly, responsive, and capable of presenting healthcare data in a visually meaningful format. Additionally, the system should ensure accurate data processing, smooth dashboard performance, and easy accessibility through a web browser. These requirements provide the foundation for developing an efficient healthcare analytics solution that supports data-driven decision-making and promotes preventive healthcare.

3.1 Solution Requirement

The proposed solution is to develop a **Heart Disease Diagnostic Dashboard** that enables efficient analysis and visualization of heart disease patient data using **MySQL**, **Tableau**, and **Flask**. The system is designed to collect, store, process, and visualize healthcare data in an interactive manner, allowing users to identify important risk factors and trends associated with heart disease. The solution provides healthcare professionals, researchers, and policymakers with an easy-to-use platform for exploring patient information and making data-driven decisions.



3.2 Data Flow Diagram

The Data Flow Diagram (DFD) illustrates the movement of data through the Heart Disease Diagnostic Dashboard system. The process begins with the heart disease dataset, which contains patient information such as age, gender, cholesterol levels, blood pressure, chest pain type, exercise-induced angina, and heart disease status. The dataset is imported into a **MySQL** database, where it is stored, managed, and processed using SQL queries.

The processed data is then connected to **Tableau Public**, which generates interactive visualizations, dashboards, and stories to analyze patient health indicators and identify important disease patterns. Finally, the Tableau dashboard is embedded into a **Flask** web application, providing a user-friendly interface through which doctors, healthcare professionals, researchers, and other users can access and interact with the dashboard. This flow ensures efficient data management, meaningful visualization, and easy accessibility for better healthcare decision-making.

Data Flow Components

Input

- Heart Disease Dataset (CSV)

Process 1

- Data Storage and Management (MySQL)

Process 2

- SQL Operations (Filtering, Querying, Aggregation)

Process 3

- Data Visualization (Tableau Dashboard & Story)

Process 4

- Web Integration (Flask Application)

Output

- Interactive Dashboard
- Healthcare Insights
- Risk Factor Analysis

3.3 Technology Stack

The Heart Disease Diagnostic Dashboard is developed using a combination of database management, business intelligence, web development, and visualization technologies. Each technology plays an important role in ensuring efficient data storage, analysis, visualization, and deployment.

Technology	Purpose
Python	Used for developing the Flask web application and integrating the Tableau dashboard.
Flask	Lightweight web framework used to create the project's user interface and embed the Tableau dashboard.
MySQL	Database management system used to store, manage, and retrieve heart disease patient data using SQL queries.
SQL	Used to perform data manipulation, filtering, aggregation, and retrieval from the MySQL database.
Tableau Public	Business Intelligence (BI) tool used to create interactive visualizations, dashboards, and stories.
HTML5	Used to design the structure of the web pages.
CSS3	Used to style the web application and improve the user interface.
Git	Version control system used to track project changes.
GitHub	Repository hosting platform used for project source code management and collaboration.
Visual Studio Code	Code editor used for developing the Flask application and project files.
MySQL Workbench	GUI tool used for creating databases, importing datasets, and executing SQL queries.
Tableau Public Desktop	Desktop application used to design dashboards and publish them online.

4. PROJECT DESIGN

The **Project Design** phase defines the overall architecture, workflow, and implementation strategy of the **Heart Disease Diagnostic Dashboard**. The system is designed using a layered architecture where each component performs a specific function, ensuring efficient data processing, visualization, and user interaction. The project begins with collecting the heart disease dataset, which is imported into a **MySQL** database. SQL queries are then used to clean, organize, and retrieve the required data for analysis. The processed data is connected to **Tableau Public**, where interactive charts, dashboards, and stories are created to visualize important health indicators and disease trends. Finally, the Tableau dashboard is embedded into a **Flask** web application, allowing users to access the dashboard through a simple and responsive web interface.

The project design follows a modular approach, making the system easy to maintain and extend. Each module—data storage, data processing, visualization, and web integration—works independently while interacting seamlessly with the others. This architecture improves scalability, reduces complexity, and ensures that users can explore healthcare data efficiently through interactive visualizations. The overall design enables healthcare professionals, researchers, and decision-makers to gain meaningful insights from patient data and supports data-driven healthcare analysis.

Project Workflow

1. Collect the Heart Disease dataset (CSV).
2. Import the dataset into the MySQL database.
3. Perform SQL queries for data management and analysis.
4. Connect MySQL with Tableau Public.
5. Create interactive charts, dashboards, and stories in Tableau.
6. Publish the Tableau dashboard to Tableau Public.
7. Embed the published dashboard into a Flask web application.
8. Display the dashboard to users through a web browser for interactive analysis.

4.1 Problem Solution Fit

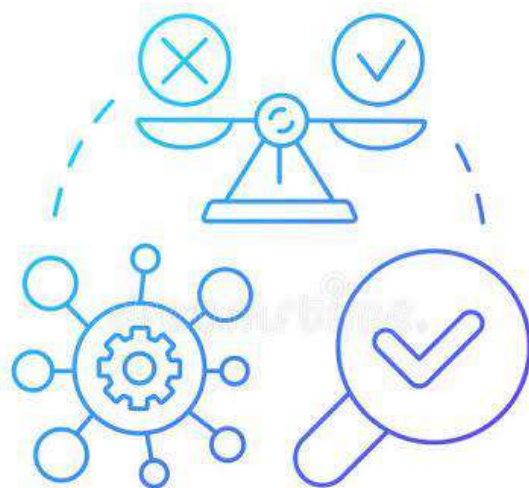
The **Heart Disease Diagnostic Dashboard** is designed to address the challenges of analyzing large volumes of heart disease patient data and presenting the results in a clear and interactive format. Traditional methods of reviewing medical records and reports are often time-consuming and make it difficult to identify patterns, trends, and high-risk patient groups. This project provides an efficient solution by integrating **MySQL**, **Tableau**, and **Flask** to automate data storage, analysis, visualization, and presentation.

The solution effectively meets the identified problem by storing patient data in a **MySQL** database, where SQL queries are used to retrieve and process the required information. The processed data is visualized using **Tableau Public** through interactive charts, dashboards, and stories that display key health indicators such as age, gender, cholesterol levels, blood pressure, chest pain type, and heart disease status. The dashboard is embedded into a **Flask** web application, enabling users to access and interact with the analysis through a web browser. This integrated approach helps healthcare professionals, researchers, and policymakers make informed decisions, identify risk factors, and support preventive healthcare initiatives through data-driven insights.

4.2 Proposed Solution

The proposed solution is to develop a **Heart Disease Diagnostic Dashboard** that provides an interactive platform for analyzing and visualizing heart disease patient data. The system utilizes **MySQL** as the database to store and manage patient records efficiently. SQL queries are used to retrieve, filter, and analyze the data, ensuring accurate and organized information for visualization. The processed data is then connected to **Tableau Public**, where interactive charts, dashboards, and stories are created to present key health indicators such as age, gender, cholesterol levels, blood pressure, chest pain type, exercise-induced angina, and heart disease status.

To improve accessibility and user experience, the Tableau dashboard is embedded into a **Flask** web application using HTML and CSS. The web interface allows users to explore the dashboard through interactive filters and visualizations, making it easier to identify trends, compare patient groups, and understand the factors contributing to heart disease. This integrated solution combines database management, data analytics, business intelligence, and web technologies to deliver meaningful insights that support healthcare professionals, researchers, and policymakers in making informed, data-driven decisions and promoting preventive healthcare.



PROPOSED SOLUTION

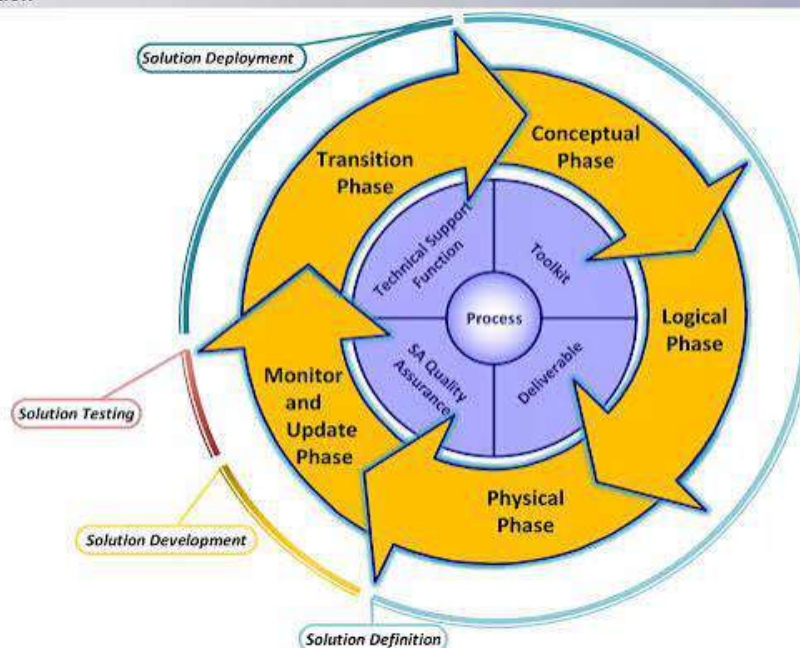
4.3 Solution Architecture

The **Solution Architecture** of the Heart Disease Diagnostic Dashboard follows a layered architecture that integrates **MySQL**, **Tableau**, and **Flask** to provide an efficient healthcare analytics solution. The architecture begins with the heart disease dataset, which is imported into a MySQL database for secure storage and management. SQL queries are used to retrieve, filter, and process the required patient data. The processed data is then connected to Tableau Public, where interactive charts, dashboards, and stories are developed to visualize important health indicators and disease trends. Finally, the Tableau dashboard is embedded into a Flask web application, enabling users to access the visualizations through a web browser.

The architecture ensures smooth data flow between all components, allowing efficient data management, interactive visualization, and easy accessibility. The modular design improves scalability, maintainability, and performance by separating data storage, processing, visualization, and presentation into independent layers. This architecture enables healthcare professionals, researchers, and policymakers to analyze patient information effectively and make informed, data-driven decisions.

Solution Architecture Framework (SAF)

Definition



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The **Project Planning** phase involved defining the project objectives, selecting the appropriate technologies, allocating tasks, and establishing a structured workflow for successful implementation. The project was planned to be completed in sequential phases, starting with data collection and database creation, followed by data analysis, visualization, web integration, testing, and deployment. A systematic approach ensured that each phase was completed before moving to the next, reducing errors and improving the overall quality of the project.

The project utilized **MySQL** for database management, **Tableau Public** for creating interactive dashboards and stories, and **Flask** for integrating the dashboard into a web application. During the planning phase, responsibilities were assigned for dataset preparation, SQL operations, dashboard development, web integration, testing, documentation, and project presentation. Regular verification of each module ensured smooth integration and timely completion of the project.

Phase	Activity
Phase 1	Problem Identification and Requirement Analysis
Phase 2	Dataset Collection and Data Understanding
Phase 3	Database Creation and Data Import using MySQL
Phase 4	SQL Operations and Data Preparation
Phase 5	Data Visualization using Tableau Public
Phase 6	Dashboard and Story Creation
Phase 7	Flask Web Application Development
Phase 8	Dashboard Integration and Testing
Phase 9	GitHub Deployment and Documentation
Phase 10	Project Documentation & Final Submission

6. Functional and Performance Testing

6.1 Functional Testing

It will explain testing of:

- Dataset Import
- SQL Query Execution
- Data Cleaning
- Dashboard Visualizations
- Interactive Filters
- Story Navigation
- Flask Integration
- Dashboard Responsiveness

Functional Testing Table

Test ID	Module Test Case		Expected Result	Actual Result	Status
FT01	MySQL	Import CSV	Dataset imported successfully	Successful	Pass
FT02	SQL	Execute Queries	Correct records displayed	Successful	Pass
FT03	Tableau	Charts Generated	Charts displayed correctly	Successful	Pass
FT04	Filters	Gender Filter	Dashboard updates instantly	Successful	Pass
FT05	Story	Story Navigation	Story pages open correctly	Successful	Pass
FT06	Flask	Dashboard Embedded	Dashboard loads in browser	Successful	Pass

6.2 Performance Testing

Professional explanation covering

- Dashboard loading speed
- Response time
- Query execution
- Memory utilization
- Browser compatibility
- Filter performance
- Visualization rendering

Performance Testing Table

Parameter	Description	Result
Dataset Loading	303 records loaded	Successful
SQL Execution	Queries executed correctly	Pass
Dashboard Load Time	Dashboard loaded smoothly	Pass
Filter Response	Filters responded instantly	Pass
Chart Rendering	All charts displayed	Pass
Story Navigation	Story opened correctly	Pass
Flask Integration	Embedded dashboard working	Pass
Browser Compatibility	Chrome, Edge tested	Pass

7. RESULTS

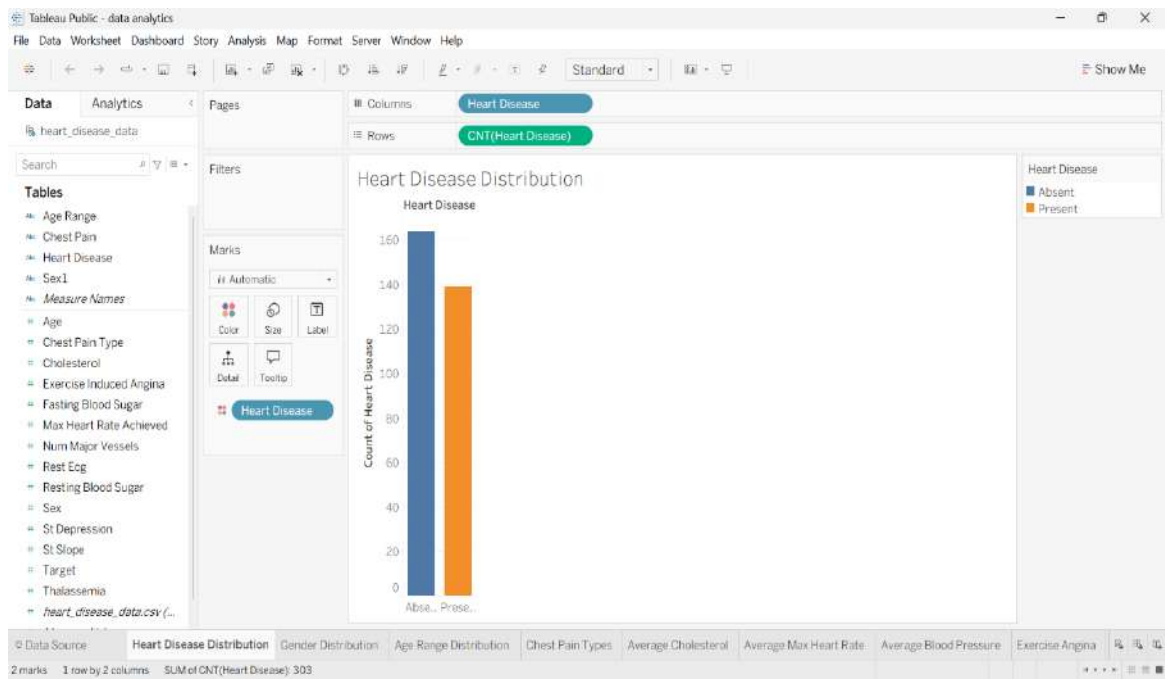
The Heart Disease Analysis Dashboard was successfully developed by integrating MySQL, SQL, Tableau Public, and Flask. The project transformed raw heart disease patient data into interactive dashboards and story points that provide meaningful insights into various clinical and demographic factors associated with heart disease.

The dataset was first stored and managed in a MySQL database, where SQL queries were used to clean, organize, and retrieve relevant patient information. The processed data was then connected to Tableau Public to create interactive worksheets, dashboards, and stories. Finally, the dashboard was embedded into a Flask web application, allowing users to access the visualizations through a web browser.

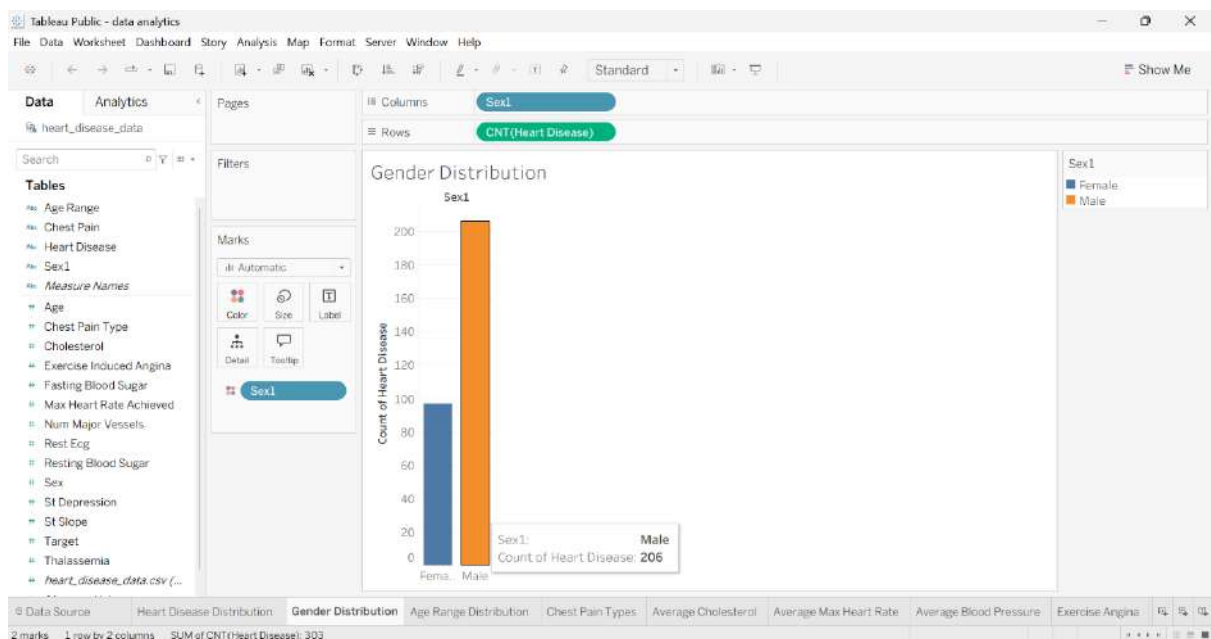
The results demonstrate how interactive data visualization simplifies complex healthcare data and helps users identify disease patterns, compare patient characteristics, and support data-driven decision-making.

7.1 Worksheet Visualizations

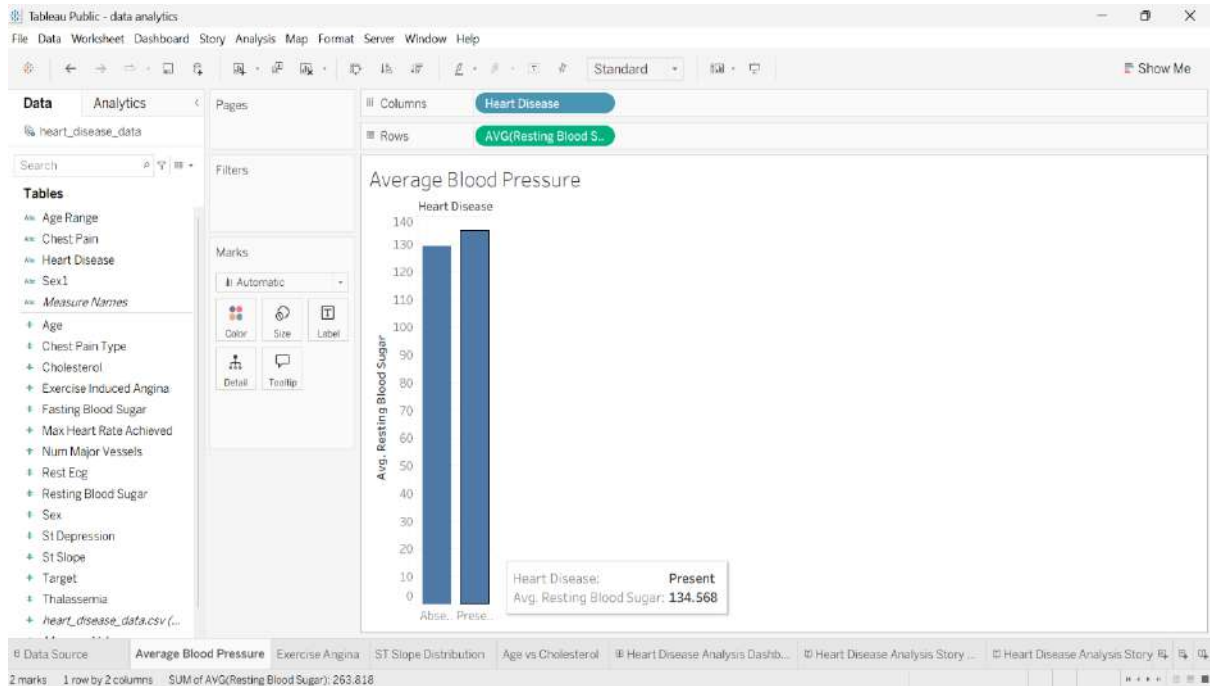
Heart Disease Distribution



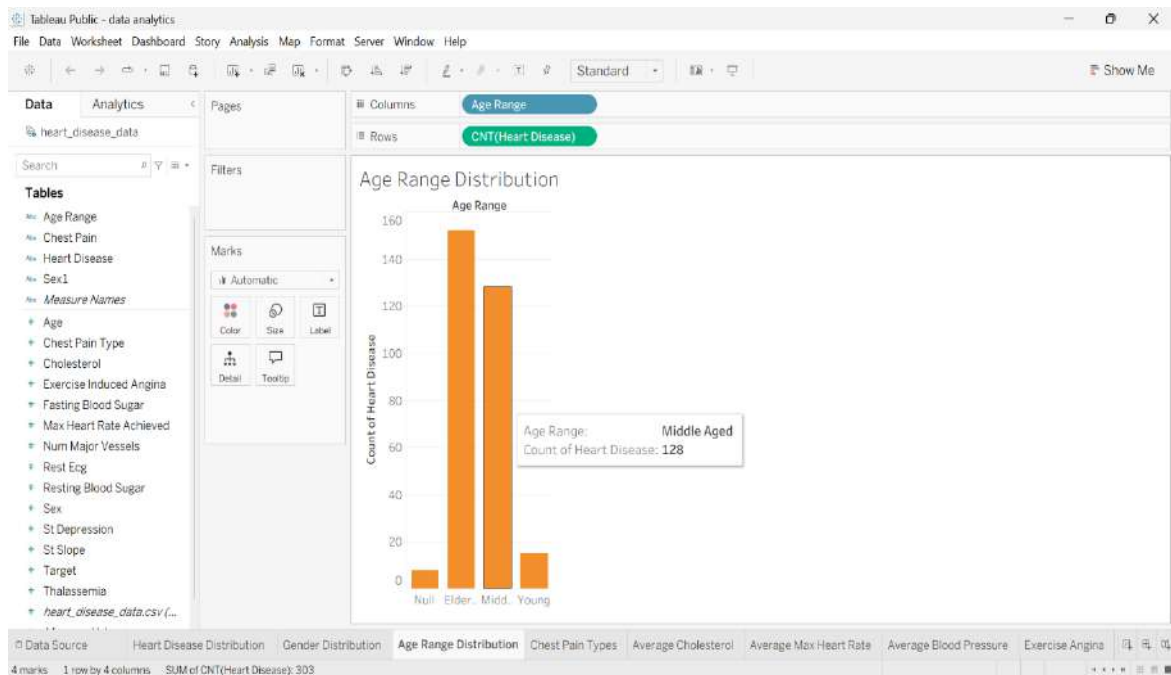
Gender Distribution



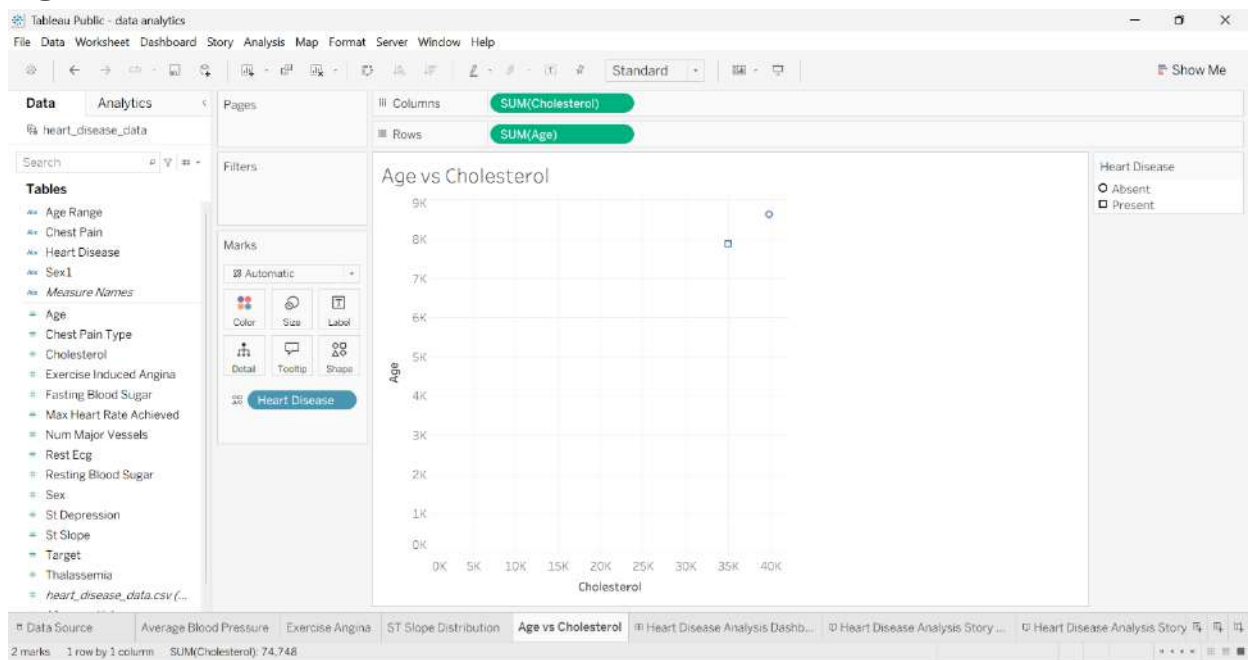
Average Blood Pressure



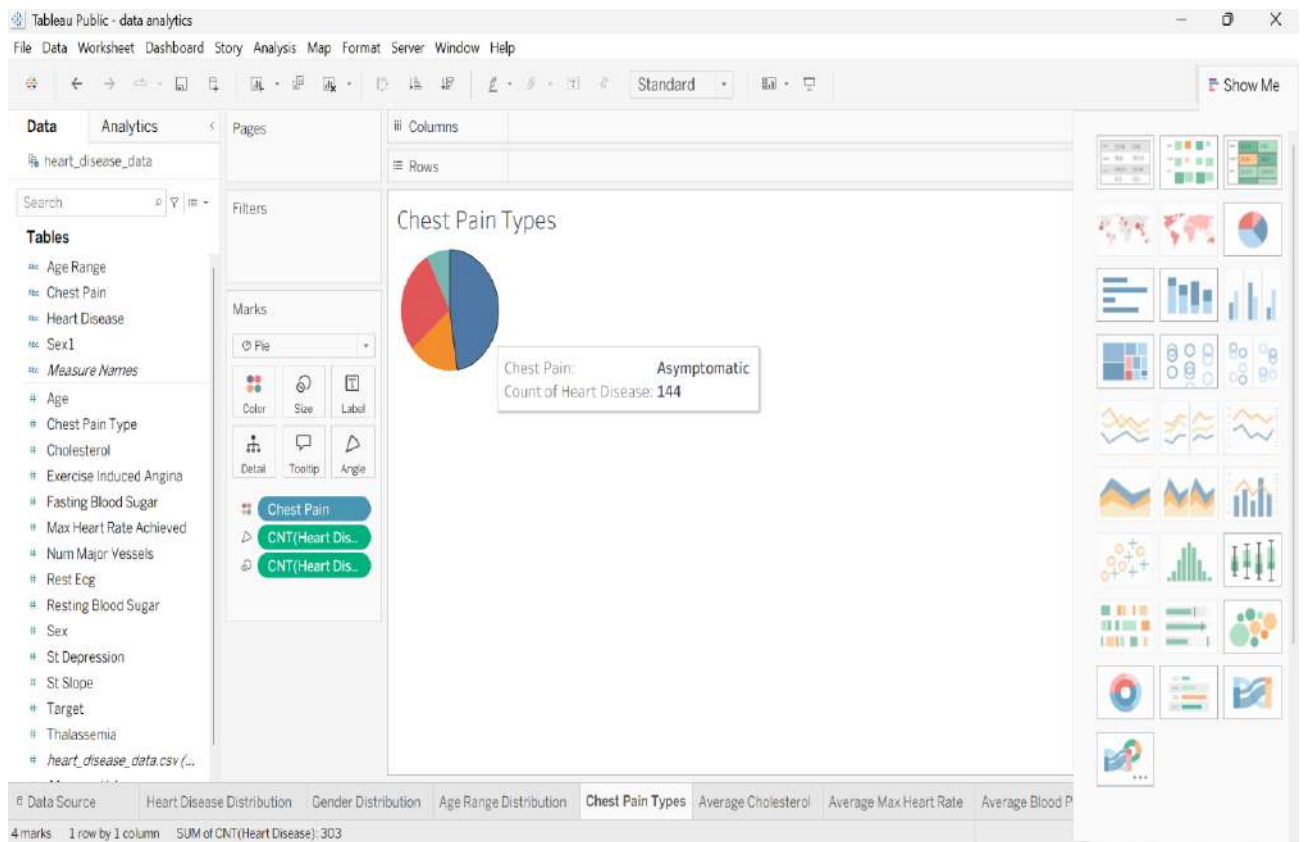
Age Range Distribution



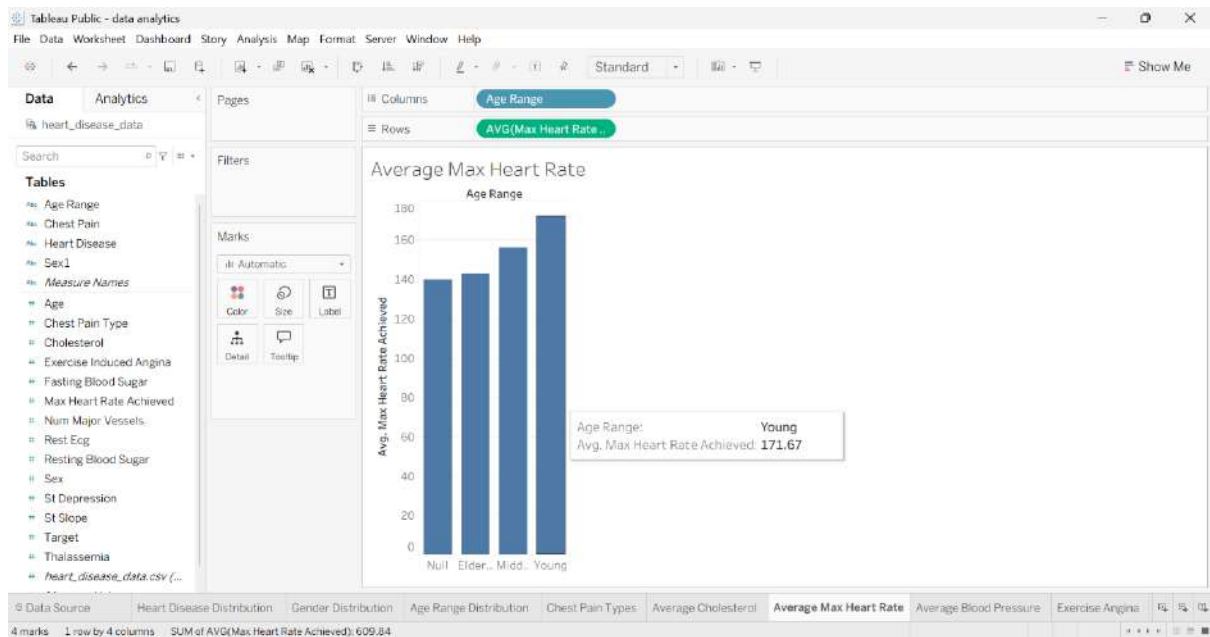
Age vs Cholesterol



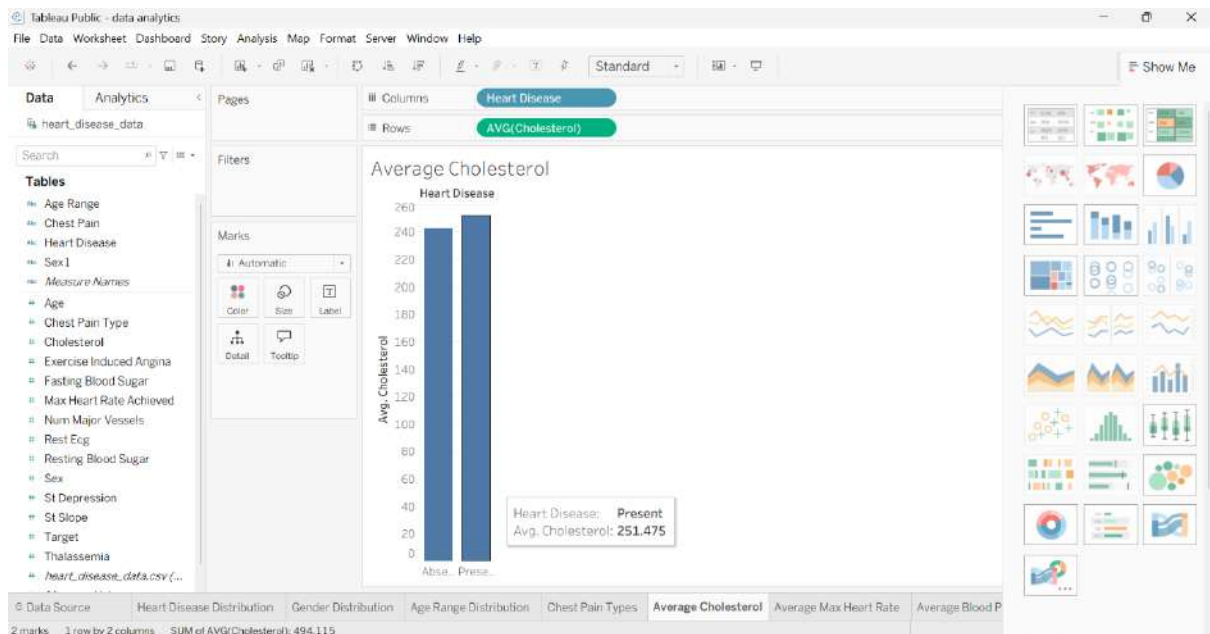
Chest pain Types



Average Max Heart Rate



Average Cholesterol



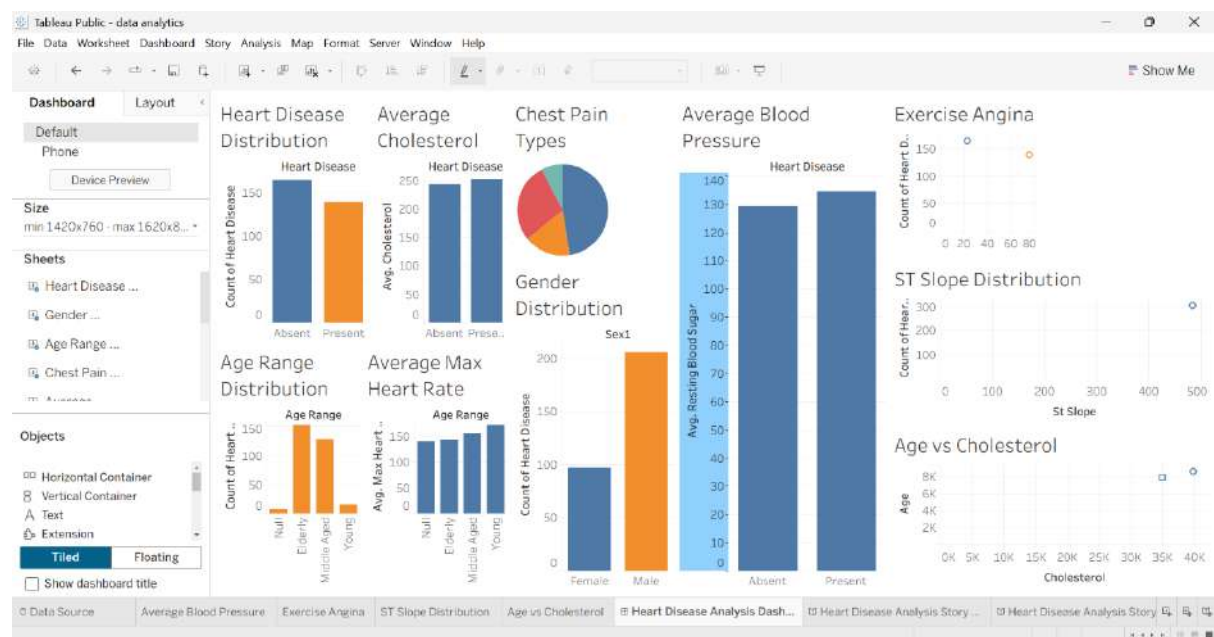
7.2 Dashboard Output

An interactive dashboard was developed by combining multiple worksheets into a single interface. The dashboard enables users to analyze patient information through interactive filters and visualizations.

The dashboard provides a comprehensive overview of heart disease indicators and allows healthcare professionals to compare different patient groups quickly. Users can interact with charts and filters to gain deeper insights into patient characteristics and disease risk factors.

Tableau Profile Link:

https://public.tableau.com/views/heartdiseasediagnosticdashboard/STSLopeDistribution?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link



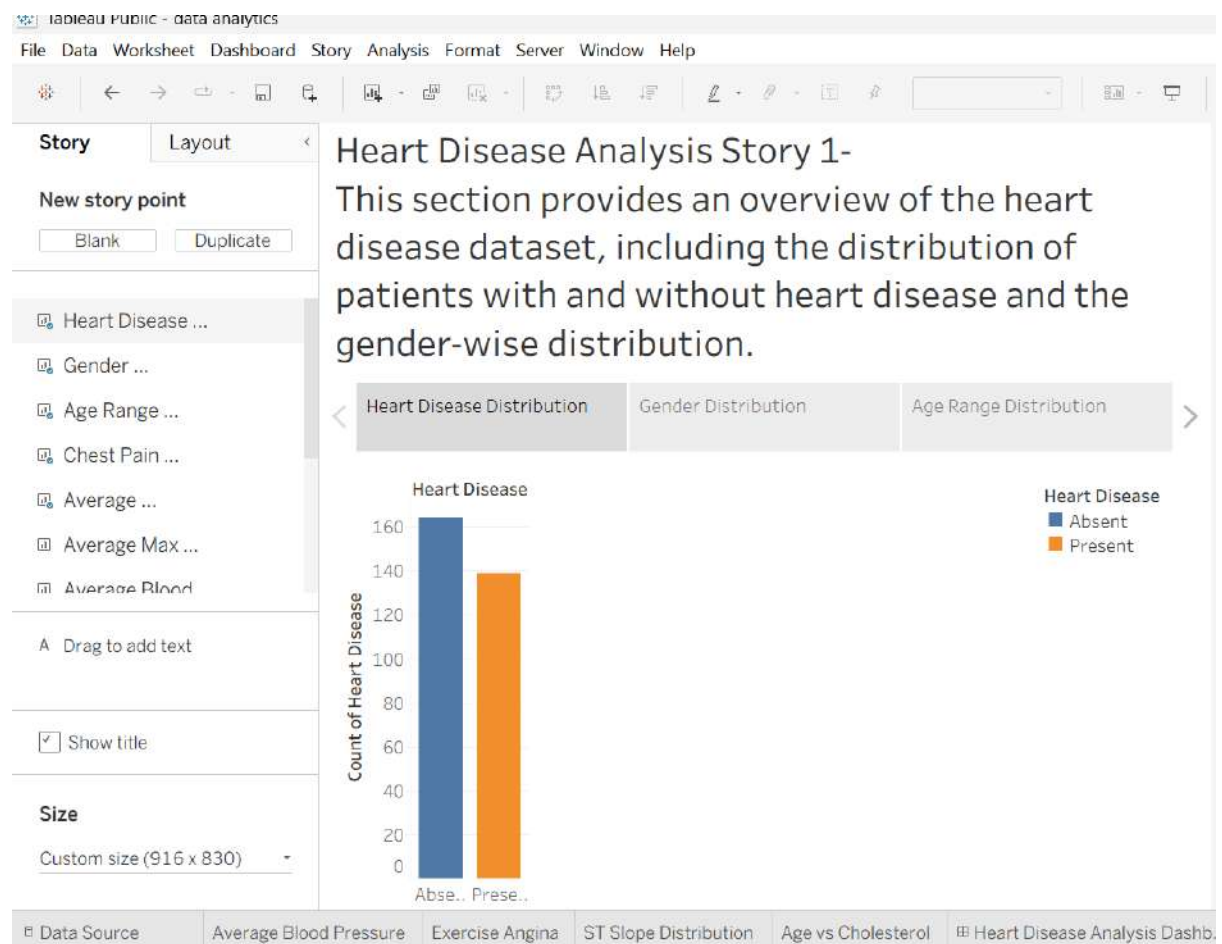
7.3 Story Output

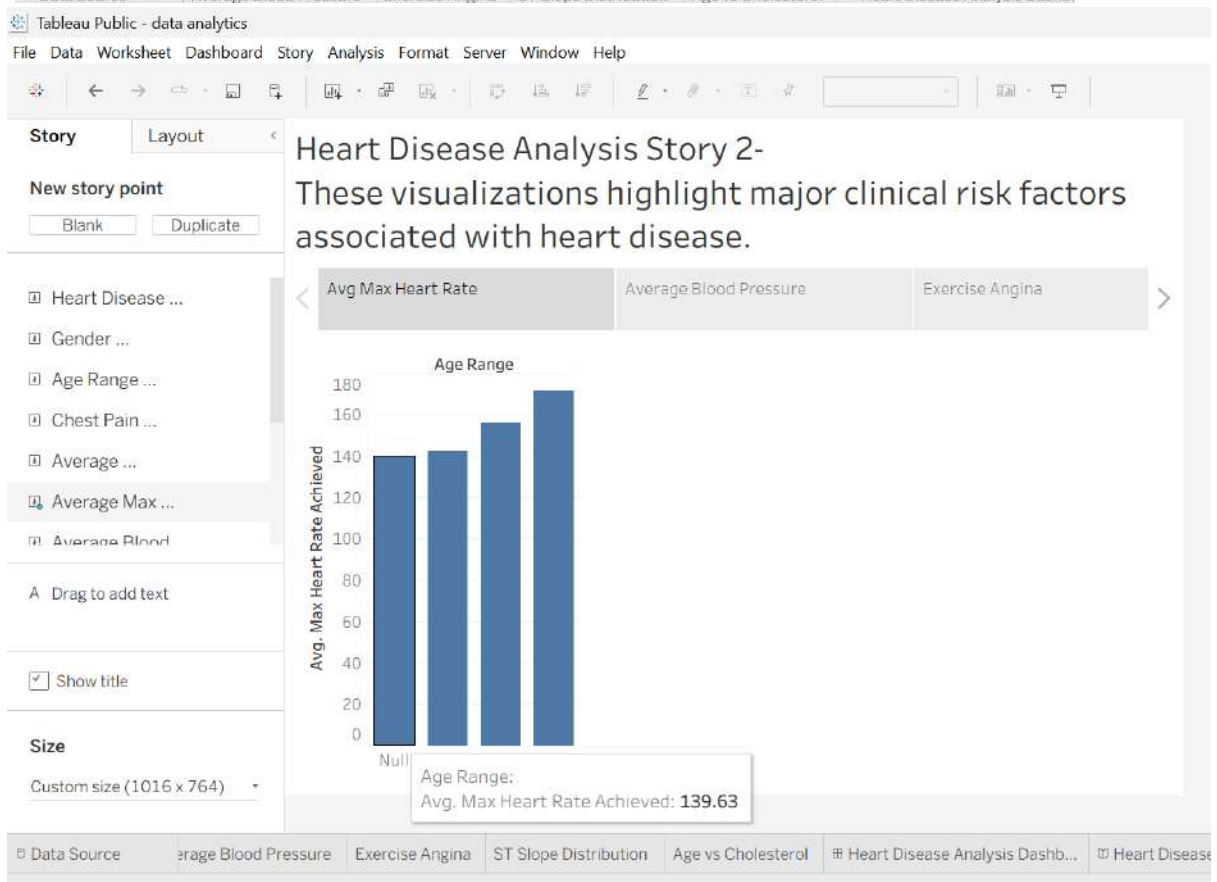
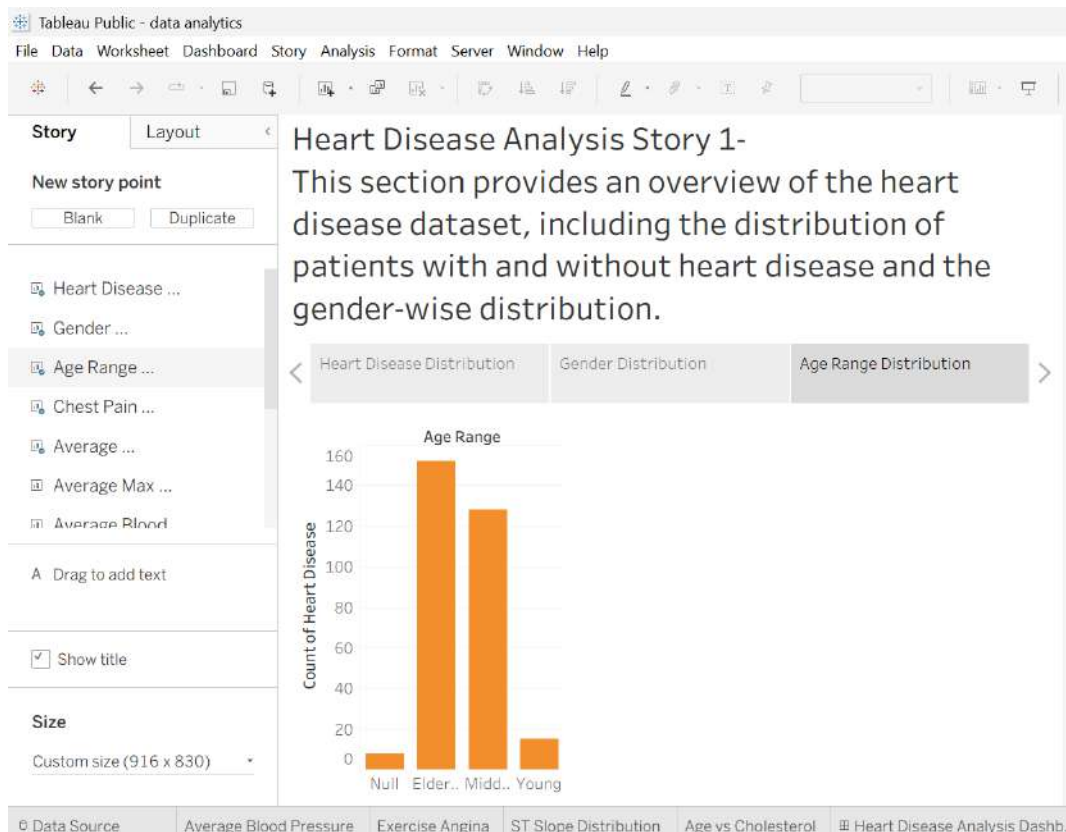
A Tableau Story was created to present the complete analysis in a logical sequence. Each story point explains one aspect of the heart disease analysis, making it easier for users to understand the insights generated from the dataset.

The story begins with an overview of the dataset, followed by analyses of gender distribution, age groups, cholesterol levels, chest pain types, exercise-induced angina, and concludes with key observations regarding heart disease risk factors.

Tableau Public Link:

https://public.tableau.com/views/heartdiseasediagnosticdashboard/STSlopeDistribution?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link





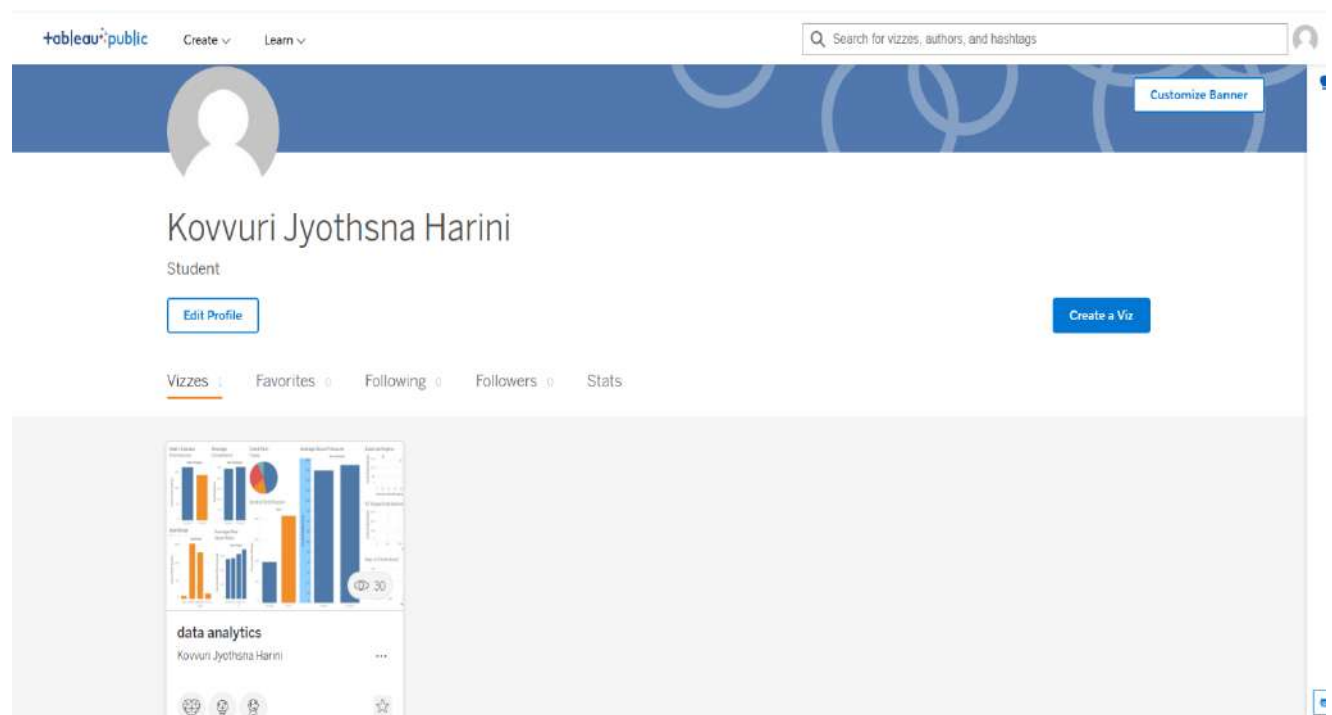
7.4 Tableau Public Output

The Tableau Dashboard was successfully embedded into a Flask web application developed using Python, HTML, and CSS. The web application provides a simple interface for users to access the dashboard directly through a browser.

The Flask application improves accessibility by integrating the published Tableau dashboard into a web environment, allowing users to interact with the visualizations without opening Tableau separately.

Tableau Public Profile Link:

<https://public.tableau.com/app/profile/kovvuri.jyothsna.harini/vizzes>



8. ADVANTAGES & DISADVANTAGES

Introduction

Every data analytics project has both strengths and limitations. The Heart Disease Analysis Dashboard was developed to transform raw healthcare data into meaningful visual insights using MySQL, SQL, Tableau Public, and Flask. The project enables healthcare professionals, researchers, and decision-makers to analyze patient information through interactive dashboards and visualizations. While the system provides an efficient platform for exploring heart disease patterns and risk factors, its performance depends on the quality of the available dataset and the capabilities of the underlying technologies. Understanding both the advantages and limitations helps evaluate the effectiveness of the system and identify opportunities for future improvements.

Advantages

- Converts raw heart disease patient data into meaningful and interactive visualizations. Enables quick identification of major heart disease risk factors such as age, cholesterol, blood pressure, chest pain type, and exercise-induced angina.
- Provides an interactive dashboard with filters that allow users to explore different patient groups.
- Supports data-driven decision-making for healthcare professionals and researchers.
- Integrates MySQL, SQL, Tableau Public, and Flask into a complete analytics solution.
- Reduces the time required to analyze large volumes of healthcare data.
- Easy to maintain and update when new patient datasets become available.

Disadvantages

- The analysis depends entirely on the accuracy and completeness of the available dataset.
- The dashboard performs descriptive analysis only and does not predict future heart disease risk.
- It does not support real-time patient data updates.
- Large datasets may increase dashboard loading time depending on system resources.
- The Tableau Public version has certain sharing and customization limitations compared to Tableau Server or Tableau Cloud.
- Users require internet access to view the published Tableau dashboard.

9. CONCLUSION

The **Heart Disease Analysis Dashboard** was successfully developed to transform raw heart disease patient data into meaningful and interactive visualizations using MySQL, SQL, Tableau Public, Flask, Python, HTML, and CSS. The project demonstrates how Business Intelligence and Data Analytics techniques can simplify the analysis of complex healthcare datasets and present valuable insights in an easy-to-understand format.

The project began with collecting the heart disease dataset and storing it in a MySQL database. SQL queries were used to clean, organize, retrieve, and analyze the data efficiently. The processed data was then connected to Tableau Public, where interactive worksheets, dashboards, and story points were created to visualize important clinical indicators such as age, gender, cholesterol levels, resting blood pressure, chest pain type, exercise-induced angina, maximum heart rate, and heart disease status. Finally, the Tableau dashboard was integrated into a Flask web application, allowing users to access the analysis through a simple and responsive web interface.

The developed dashboard enables healthcare professionals, researchers, and decision-makers to explore patient information through interactive filters and visual reports. It helps identify disease patterns, compare patient groups, and understand the relationships between different health indicators. By presenting healthcare data through clear visualizations, the system supports faster interpretation of information and promotes informed, data-driven decision-making.

10. FUTURE SCOPE

The Heart Disease Analysis Dashboard provides an effective platform for analyzing patient data through interactive visualizations. Although the current system successfully supports descriptive analysis using MySQL, SQL, Tableau Public, and Flask, there are several opportunities to enhance its functionality and make it more intelligent, scalable, and suitable for real-world healthcare applications.

Furthermore, stronger security mechanisms such as user authentication, role-based access control, encrypted data storage, and secure communication protocols can be implemented to protect sensitive patient information and comply with healthcare data privacy standards.

Overall, these future enhancements would transform the current Heart Disease Analysis Dashboard into a comprehensive, intelligent, and scalable healthcare analytics platform capable of supporting predictive analysis, real-time monitoring, and advanced clinical decision-making.

11.APPENDIX

Source Code

The Heart Disease Analysis project was developed using Python, Flask, MySQL, SQL, Tableau Public, HTML, and CSS. The source code includes the Flask application, SQL scripts, Tableau workbook, HTML templates, CSS files, and supporting Python modules required to build and deploy the project.



Dataset Link:

<https://www.kaggle.com/code/rawanyasser42x/heart-disease-analysis>



GitHub & Project Demo Link

GitHub Repository:

<https://github.com/likithasatti/heart-disease-analysis>

Tableau Public Link:

https://public.tableau.com/views/heartdiseasediagnosticdashboard/STSlopeDistribution?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Tableau Public Profile Link:

<https://public.tableau.com/app/profile/kovvuri.jyothsna.harini/vizzes>

Tableau Public Story:

<https://public.tableau.com/app/profile/likitha.satti/viz/heartdiseasediagnosticdashboard/STSlopeDistribution>

Project Demo Video:

<https://youtu.be/QBQRcAfQi64?si=K38-nkNcLKJ7rfB0>